

Comparison of the effectiveness of 0.5% tea, 2% neem and 0.2% chlorhexidine mouthwashes on oral health: a randomized control trial.



BACKGROUND: The aim of this study was to evaluate and compare the effectiveness of 0.5% tea, 2% neem, and 0.2% chlorhexidine mouthwashes on oral health. **MATERIALS AND METHODS:** A randomized blinded controlled trial with 30 healthy human volunteers of age group 18-25 years was carried out. The subjects were randomly assigned to 3 groups i.e., group A - 0.2% chlorhexidine gluconate (bench mark control), Group B - 2% neem, and group C - 0.5% tea of 10 subjects per group. Plaque accumulation and gingival condition were recorded using plaque index and gingival index. Oral hygiene was assessed by simplified oral hygiene index (OHIS). Salivary pH was assessed by indikrom pH strips. Plaque, gingival, and simplified OHI scores as well as salivary pH were recorded at baseline, immediately after 1st rinse, after 1 week, 2nd week, and 3rd week. The 3rd week was skipped for group A. **RESULTS:** Mean plaque and gingival scores were reduced over the 3 week trial period for experimental and control groups. Anti-plaque effectiveness was observed in all groups and the highest being in group C ($P < 0.05$). Neem and tea showed comparative effectiveness on gingiva better than chlorhexidine ($P < 0.05$). The salivary pH rise was sustained and significant in Group B and C compared to Group A. Oral hygiene improvement was better appreciated in Group B and Group C. **CONCLUSION:** The effectiveness of 0.5% tea was more compared to 2% neem and 0.2% chlorhexidine mouth rinse.